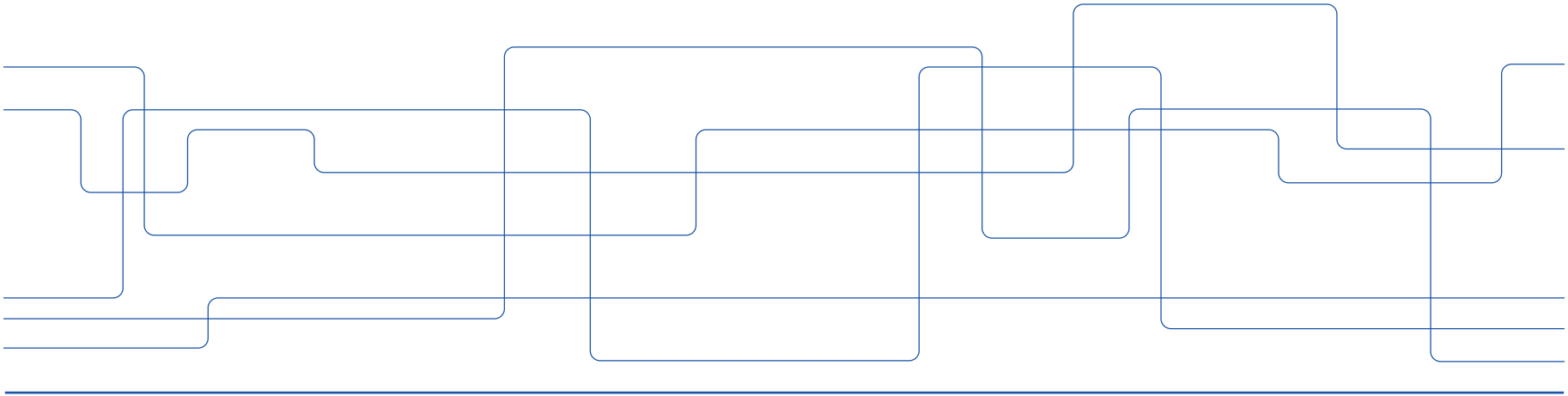
Introduction to (Practical) AI Ethics

Iolanda Leite

(Original slides from Katie Winkle)



Scope and Framing

Why this is relevant…: this talk will be completely practical (rather than philosophical), grounded mainly in empirical research and practices; covering

(1) ethical implications of AI and

(2) how to develop new technologies ethically.

Note: Will use terms AI, technology, autonomous system and robot/robotics fairly interchangeably as the distinction doesn't really matter for today's topic

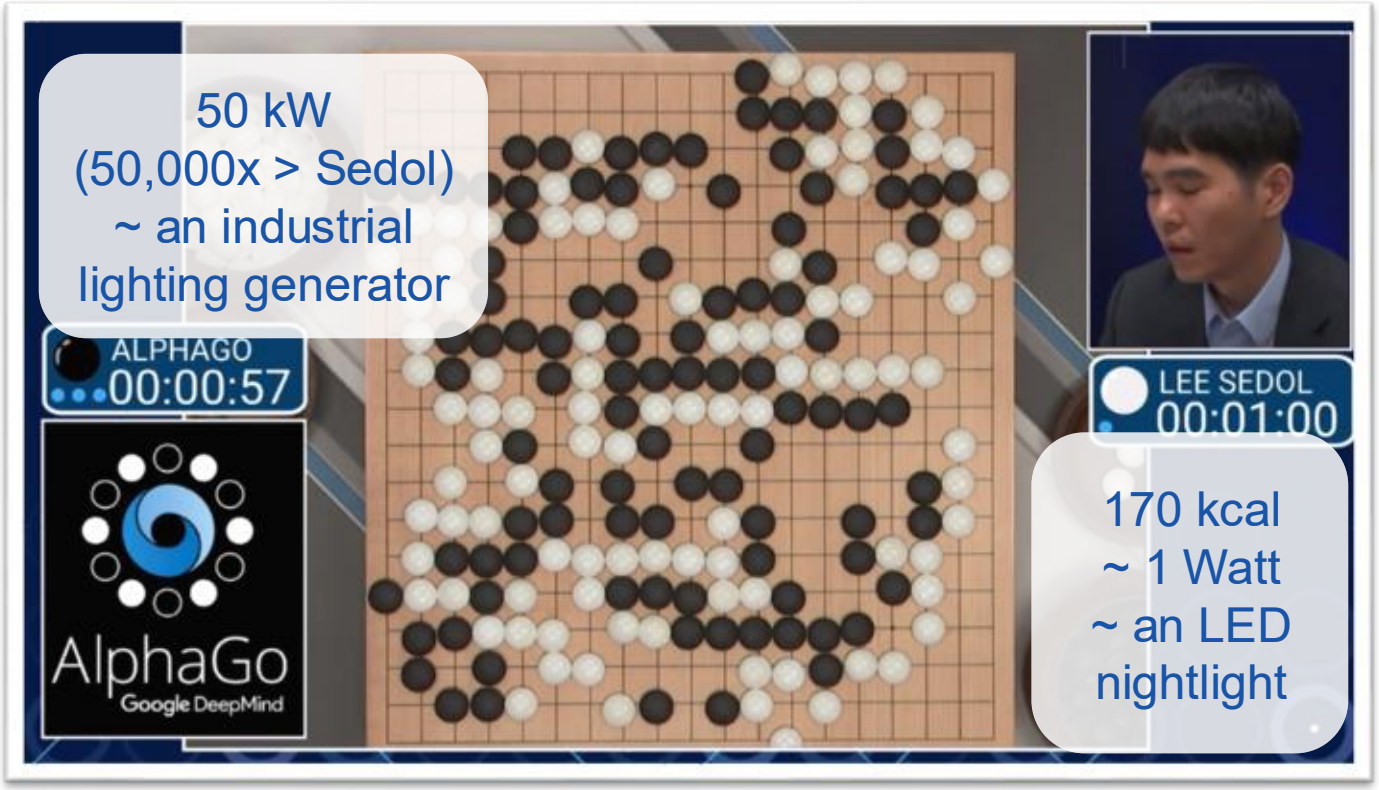
Outline

1. (Some) Topical Issues around Ethics and AI
2. Embracing Ethics as an (AI) Engineer
3. Mutually Shaping Technology and Society
4. [Break]
5. Ethical Standards in AI and Robotics
6. Worked example: Autonomous Robot Fitness Coach
7. Recap and Conclude
8. Assignment overview

Energy and Exploitation



Energy and Exploitation



Energy and Exploitation

Consumption	CO_2e (lbs)
Air travel, 1 passenger, NY↔SF	1984
Human life, avg, 1 year	11,023
American life, avg, 1 year	36,156
Car, avg incl. fuel, 1 lifetime	126,000
Training one model (GPU)	
NLP pipeline (parsing, SRL)	39
w/ tuning & experimentation	78,468
Transformer (big)	192
w/ neural architecture search	626,155

Table 1: Estimated CO_2 emissions from training common NLP models, compared to familiar consumption.[1]

Strubell, Emma, Ananya Ganesh, and Andrew McCallum. "Energy and Policy Considerations for Deep Learning in NLP." *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. 2019. [available here: https://arxiv.org/abs/1906.02243]

A more recent example:
Luccioni, Jernite & Strubell, “Power Hungry Processing: Watts Driving the Cost of AI Deployment?” Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24). https://arxiv.org/abs/2311.16863

Energy and Exploitation

Google contract worker: $15/hr = $31,200 a year



Google employee: $246,804 + the free meals, massages etc.

https://www.theguardian.com/technology/2019/may/28/a-white-collar-sweatshop-google-assistant-contractors-allege-wage-theft

Energy and Exploitation

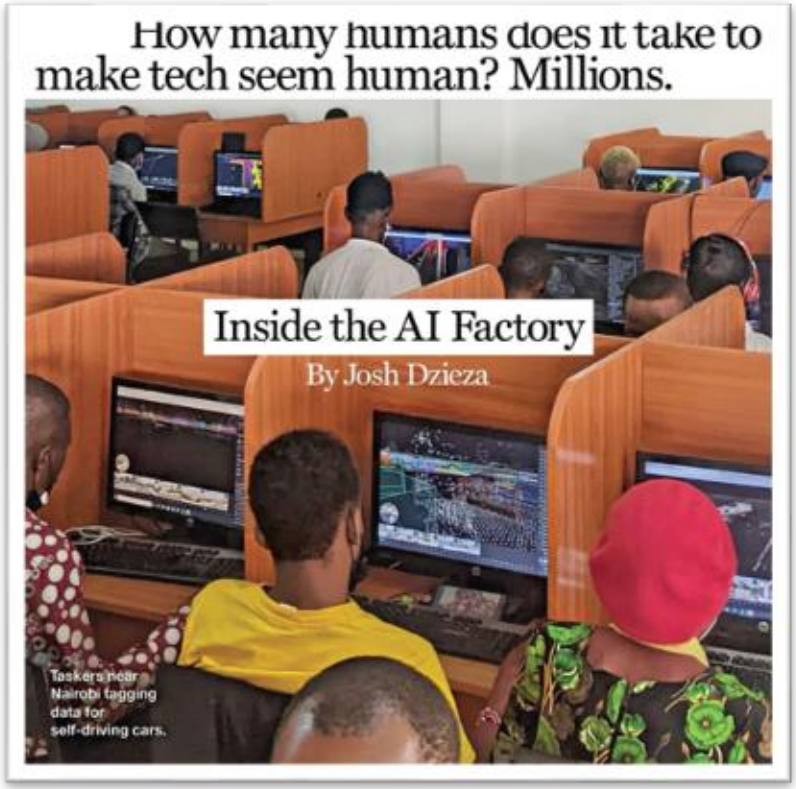


https://www.theverge.com/2019/2/25/18229714/cognizant-facebook-content-moderator-interviews-trauma-working-conditions-arizona

Energy and Exploitation

Human-in-the-loop Reinforcement Learning

"… ChatGPT seems so human because it was trained by an AI that was mimicking humans who were rating an AI that was mimicking humans who were pretending to be a better version of an AI that was trained on human writing"



https://www.theverge.com/features/23764584/ai-artificial-intelligence-data-notation-labor-scale-surge-remotasks-openai-chatbots

Energy and Exploitation



Workers wear cameras that record their movements and interactions at a factory in Karur district, Tamil Nadu. Photograph: R Satish Babu/AFP/Getty Images

When workers had cameras attached to them, they found it funny at first. But novelty soon turned to concern

https://www.theguardian.com/global-development/2026/jun/24/indian-factory-workers-told-film-themselves-for-ai-robots

Anthropomorphism and Deception



Nest Thermostat, 2011



Hello Barbie, 2015



Google Assistant, 2016



Amazon Astro, 2021

Anthropomorphism and Deception



Not every version of Barbie was a hit. Check out these flops

CNN.com
https://www.cnn.com › barbie-dolls-weird-launches

In 2015, Mattel unveiled Hello Barbie. She talked and she listened and responded to what she heard through a microphone, sophisticated electronics and a Wi-Fi connection. The doll's functions raised privacy violation concerns.

"Hello, Barbie not only listened to you, she remembered what you said, and brought it up again, sometime days later," said Richard Gottlieb, CEO of consultancy Global Toy Experts.

"It creeped parents out and earned itself a place on the cover and a scathing article in *The New York Times Magazine* about why it was not safe to have your child play with Hello Barbie," he said. "Despite, or because of making the that cover, it never took off.

Mattel discontinued the doll two years later.

https://www.nytimes.com/2015/09/20/magazine/barbie-wants-to-get-to-know-your-child.html

https://www.theguardian.com/technology/2015/nov/26/hackers-can-hijack-wi-fi-hello-barbie-to-spy-on-your-children

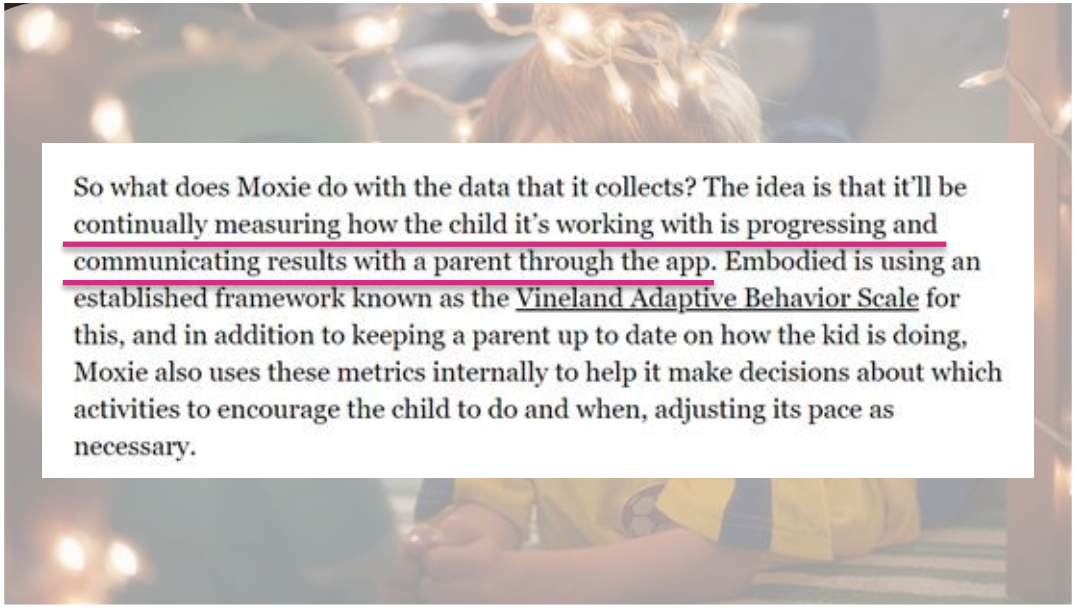
https://www.cnbc.com/2015/12/22/why-you-should-say-goodbye-to-hello-barbie.html

Anthropomorphism and Deception



https://spectrum.ieee.org/automaton/robotics/home-robots/moxie-a-social-robot-for-childhood-development

Anthropomorphism and Deception



https://spectrum.ieee.org/automaton/robotics/home-robots/moxie-a-social-robot-for-childhood-development

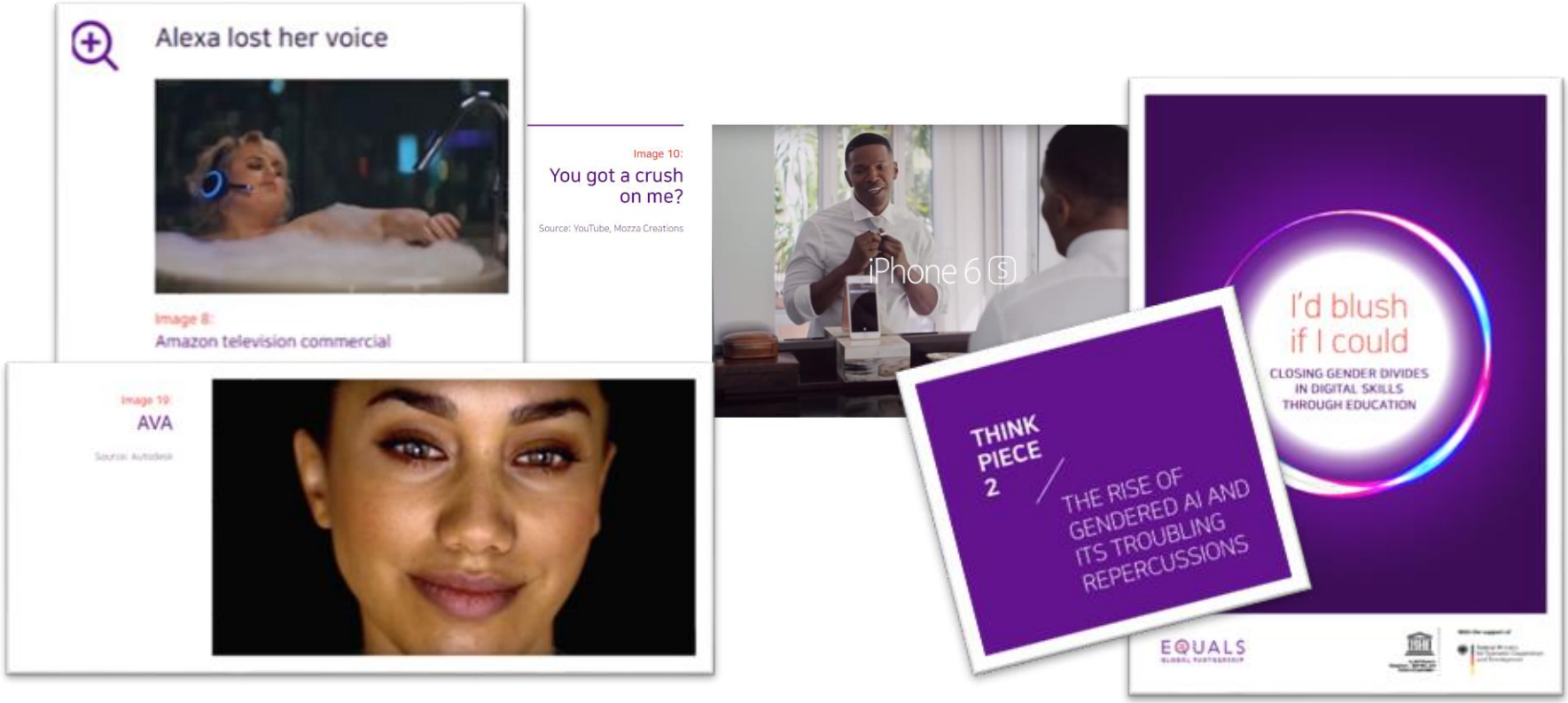
Embodied shut down in Dec 2024 — parents documented children grieving a "dying" companion, and it's now catalogued as an OECD AI incident: https://oecd.ai/en/incidents/2024-12-09-ab89

Anthropomorphism and Deception



https://youtu.be/D5VN56jQMWM

Sexism, Racism and (Mis)Representation



The Gendering of AI – UNESCO Report

Female gendered digital assistants (think Siri, Alexa) are:

- obliging, docile and eager-to-please (regardless of user tone/hostility)
- too tolerant of sexual harassment and insults, sometimes *flirting* in response to sexually explicit language

and therefore run the risk of propagating harmful stereotypes and cultural norms



Sexism, Racism and (Mis)Representation



Joy Buolamwini (as discussed in D'Ignazio and Klein's *Data Feminism*)*:*

On digging into why she wasn't recognised by facial analysis software during a class project, Buolamwini found the training data set for the facial recognition system (widely used elsewhere too) contains **78% male** and **84% white** faces.

Sexism, Racism and (Mis)Representation



In further work, on commercial gender classification systems: “*darker-skinned females are the most misclassified group (with error rates of up to 34.7%). The maximum error rate for lighter-skinned males is 0.8%”*

Buolamwini, Joy, and Timnit Gebru. "Gender shades: Intersectional accuracy disparities in commercial gender classification." *Conference on fairness, accountability and transparency*. 2018.

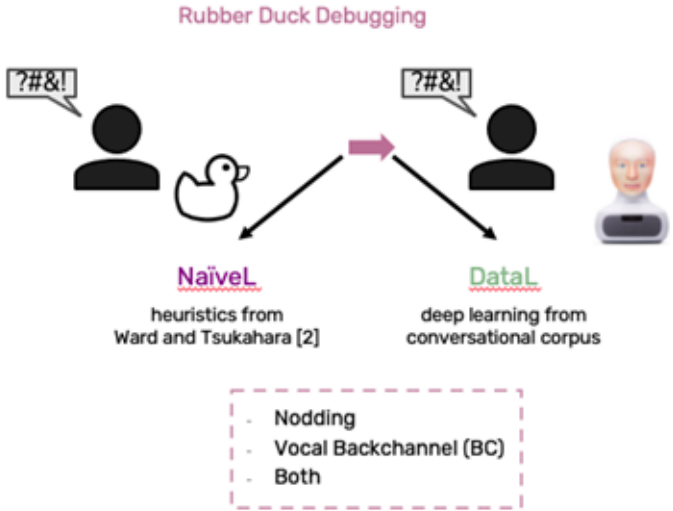
Sexism, Racism and (Mis)Representation



https://www.theguardian.com/technology/2020/sep/21/twitter-apologises-for-racist-image-cropping-algorithm



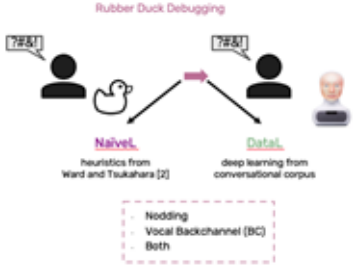
How did we miss this? A Case Study on Unintended Biases in Robot Social Behavior





Maria Teresa Parreira, Sarah Gillet, Katie Winkle, and Iolanda Leite. 2023. How Did We Miss This? A Case Study on Unintended Biases in Robot Social Behavior. In Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI '23)

How did we miss this? A Case Study on Unintended Biases in Robot Social Behavior



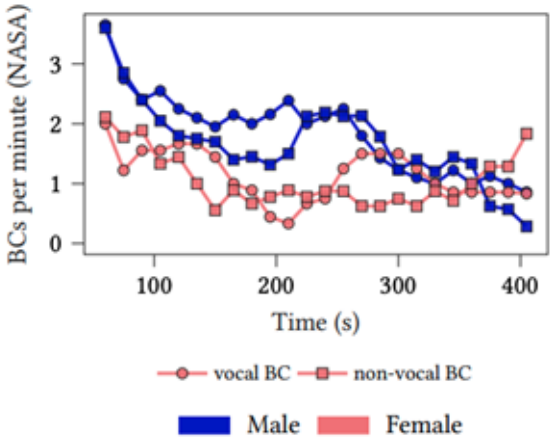
Both models backchannel ↑

with male than female participants

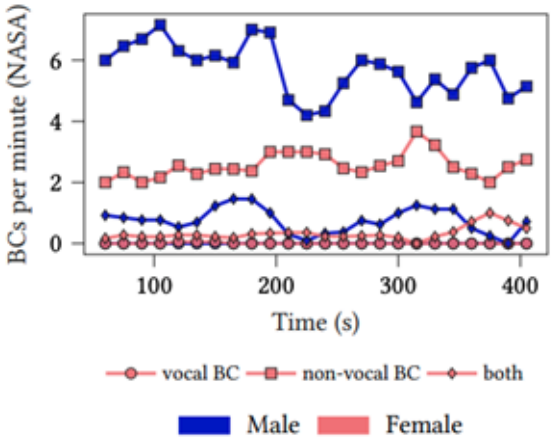
BUT

perception of the robot

did not change



Backchanneling behavior over time from the heuristic model (NaïveL) across genders (male - blue, female – red).



Backchanneling behavior over time from the data-driven model (DataL) across genders (male - blue, female – red).

Maria Teresa Parreira, Sarah Gillet, Katie Winkle, and Iolanda Leite. 2023. How Did We Miss This? A Case Study on Unintended Biases in Robot Social Behavior. In Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI '23)

Troubleshooting

Dataset Gender Imbalances

-3F, 29M - de Kok et al. [3]

-4F, 16M - Poppe et al. [4]

-67F, 37M -Morency et al. [5]

-0F, 22M - Okato et al. [6]

-2F, 10M - Ward and Tsukahara [2]

-4F, 12M - CCDb [7]

-of which we used (0F, 6M)

[2] Nigel Ward and Wataru Tsukahara. 2000. Prosodic features which cue backchannel responses in English and Japanese. Journal of Pragmatics 32, 8 1177–1207.

[3] Iwan de Kok, Dirk Heylen, and Louis-Philippe Morency. 2013. Speaker-Adaptive Multimodal Prediction Model for Listener Responses. In Proceedings of the 15th ICMI (Sydney, Australia). ACM, New York, NY, USA, 51–58

[4] Ronald Poppe, Khiet P. Truong, Dennis Reidsma, and Dirk Heylen. 2010. Backchannel Strategies for Artificial Listeners. In Proceedings of the 10th IVA (Philadelphia, PA). Springer-Verlag, Berlin, Heidelberg, 146–158

[5] Louis-Philippe Morency, Iwan Kok, and Jonathan Gratch. 2010. A probabilistic multimodal approach for predicting listener backchannels. Autonomous Agents and Multi-Agent Systems 20 (01 2010), 70–84.

[6] Y. Okato, K. Kato, M. Kamamoto, and S. Itahashi. 1996. Insertion of interjectory response based on prosodic information. In Proceedings of IVTTA '96. Workshop on Interactive Voice Technology for Telecommunications Applications. 85–88.

[7] A. J. Aubrey, D. Marshall, P. L. Rosin, J. Vandeventer, D. W. Cunningham, and C. Wallraven. 2013. Cardiff Conversation Database (CCDb): A Database of Natural Dyadic Conversations. In 2013 IEEE Conference on CVPR Workshops. 277–282.

What can be done?

Preventative measures:

- examine dataset to form a pool of gender-diverse participants
- test developed behaviors while being sensitive to impressions of the robot's behaviors across people of different identities
- quantify behavior prior to study to objectively observe the robot's behavior towards participants with different (gender) identities
- avoid over reliance on *perceptual* only; compare and contrast *objective* measures of robot

How should the robot behave? 3 different options…

- reproduce differences observed in human-human interactions
- treat users of all gender identities exactly equally
- overcompensate differences observed in human-human interactions

Copyright, fair use and creativity

The painter and illustrator Molly Crabapple is helping lead a movement of artists challenging this theft. "AI art generators are trained on enormous datasets, containing millions upon millions of copyrighted images, harvested without their creator's knowledge, let alone compensation or consent. This is effectively the greatest art heist in history. Perpetrated by respectable-seeming corporate entities backed by Silicon Valley venture capital. It's daylight robbery," a new open letter she co-drafted states.

https://www.theguardian.com/commentisfree/2023/may/08/ai-machines-hallucinating-naomi-klein

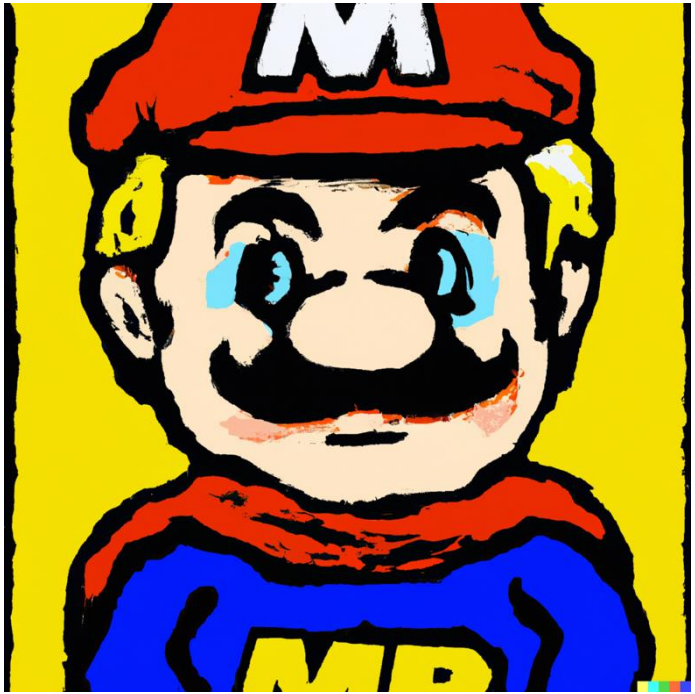


Image generated by DALL·E 2

AI-generated fake content in politics

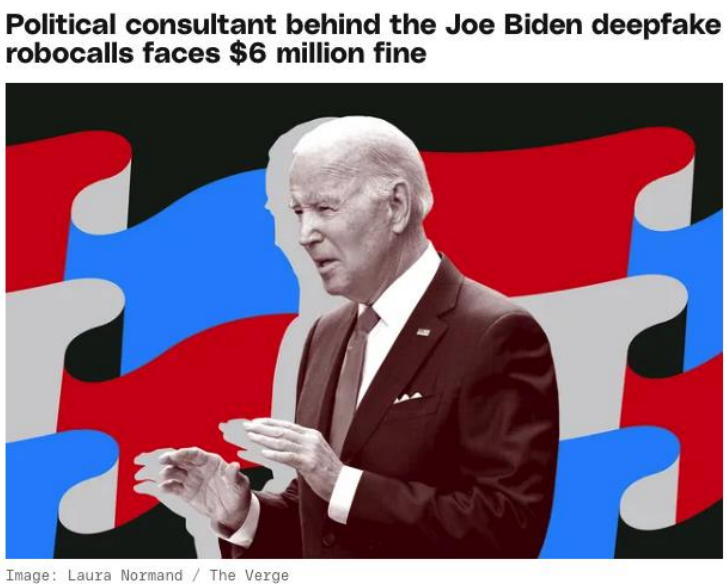
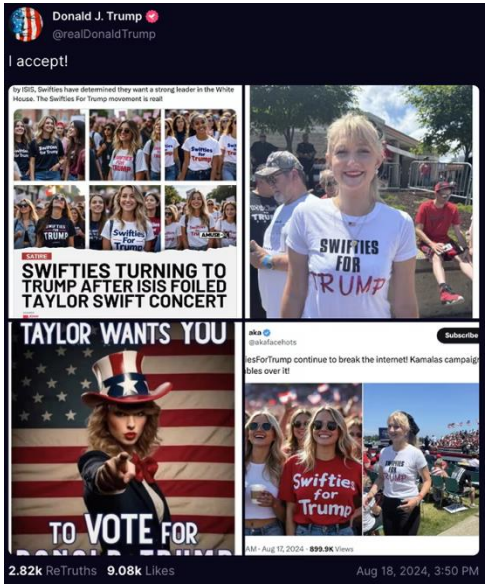


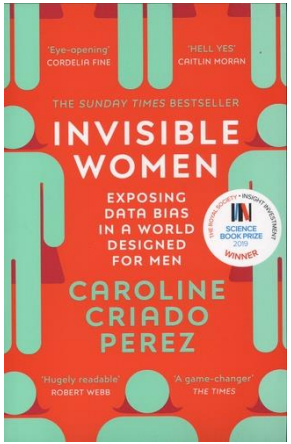
Image: Laura Normand / The Verge

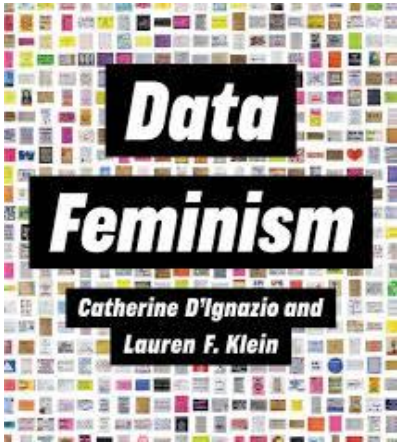


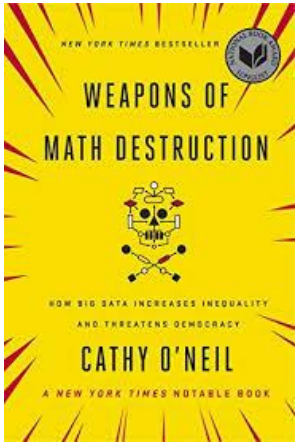


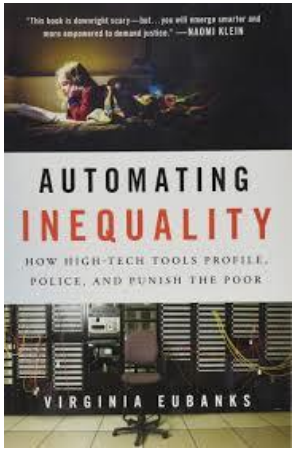
https://www.theguardian.com/us-news/article/2024/aug/19/trump-ai-swift-harris-musk-deepfake-images
https://www.bbc.com/news/world-us-canada-68440150
https://www.theverge.com/policy/24098798/2024-election-ai-generated-disinformation

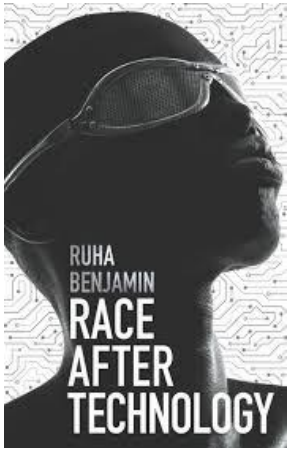
Additional suggestions: Tech + Social (In)Justice





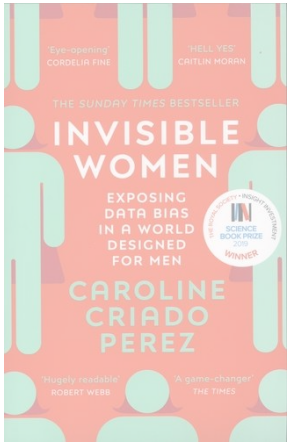






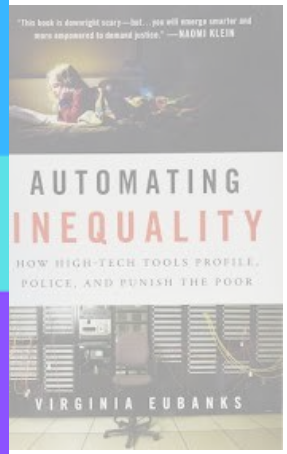
+ Ben Shneiderman notes on Human-Centred AI:
https://medium.com/@ben.shneiderman

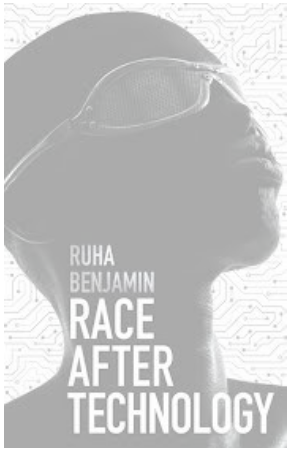
Additional suggestions: Tech + Social (In)Justice











Bonus podcast recommendation: Introduction to Feminism and Social Justice Studies ~ Dr Alex Ketchum of McGill University. Episode 17 focuses specifically on tech.

What Do I Mean By ‘Ethical AI’ ?



What Do I Mean By ‘Ethical AI’ ?

Designing and Developing (*Any and All*) AI *Ethically*

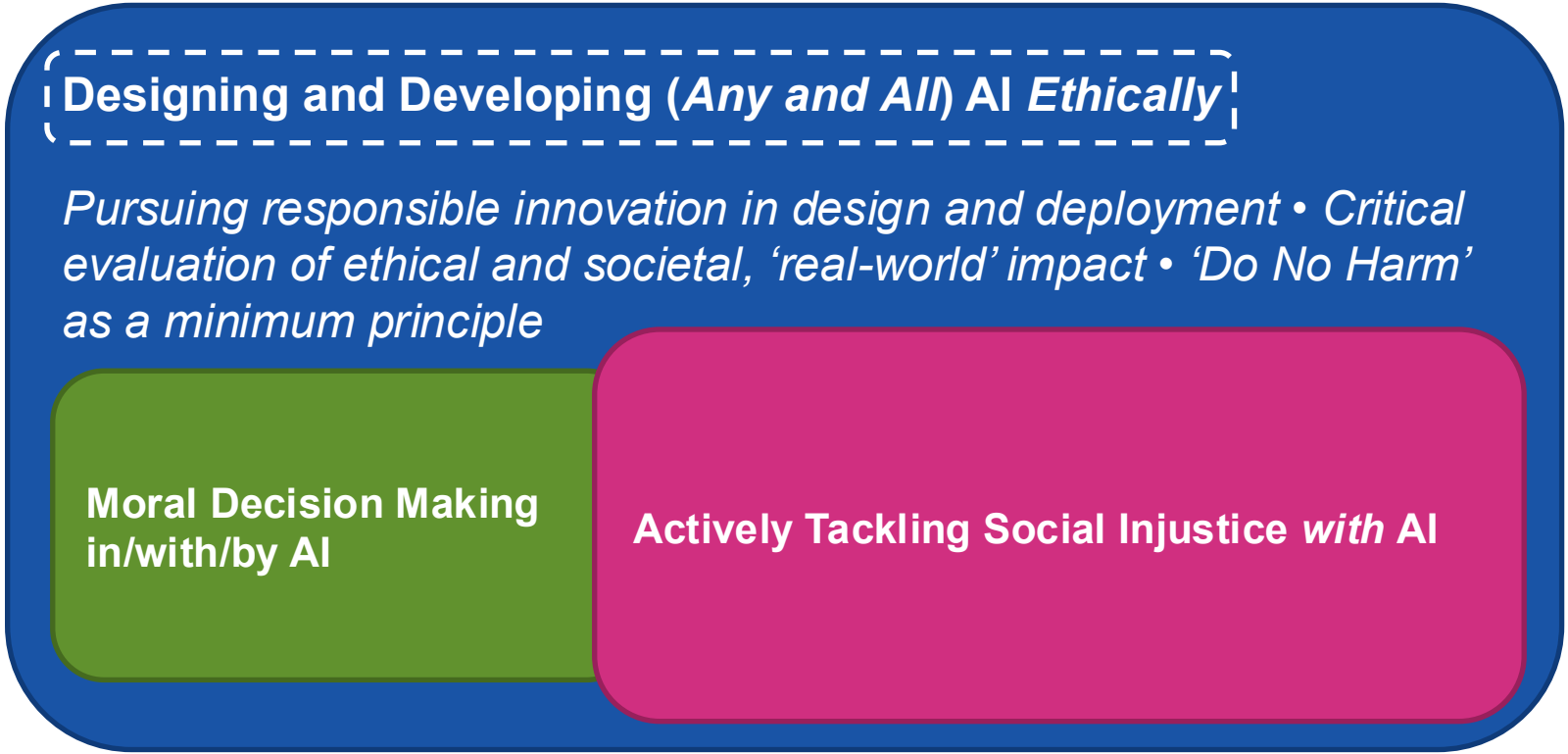
Moral Decision Making in/with/by AI

e.g. the trolley problem

Actively Tackling Social Injustice *with* AI

AI4SG Movement • *Unbiased Recruitment [AT KTH]: AI and the UN Sustainable Development Goals* • *Feminist Social Robots* • *Inclusion of Newly Arrived Children*

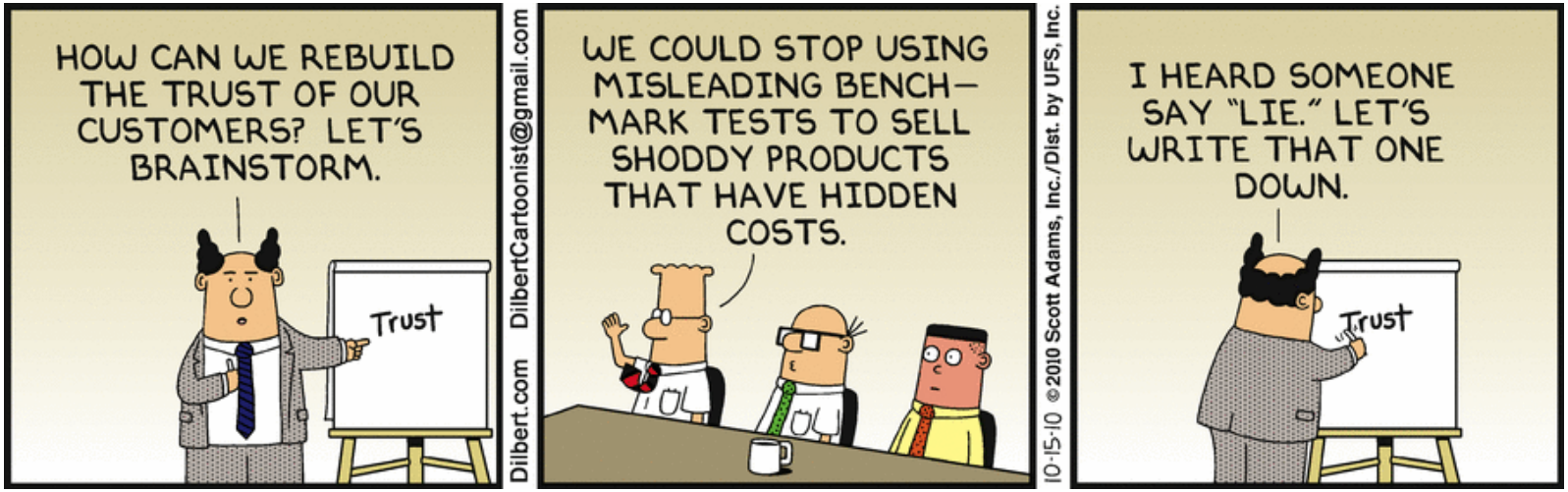
What Do I Mean By ‘Ethical AI’ ?



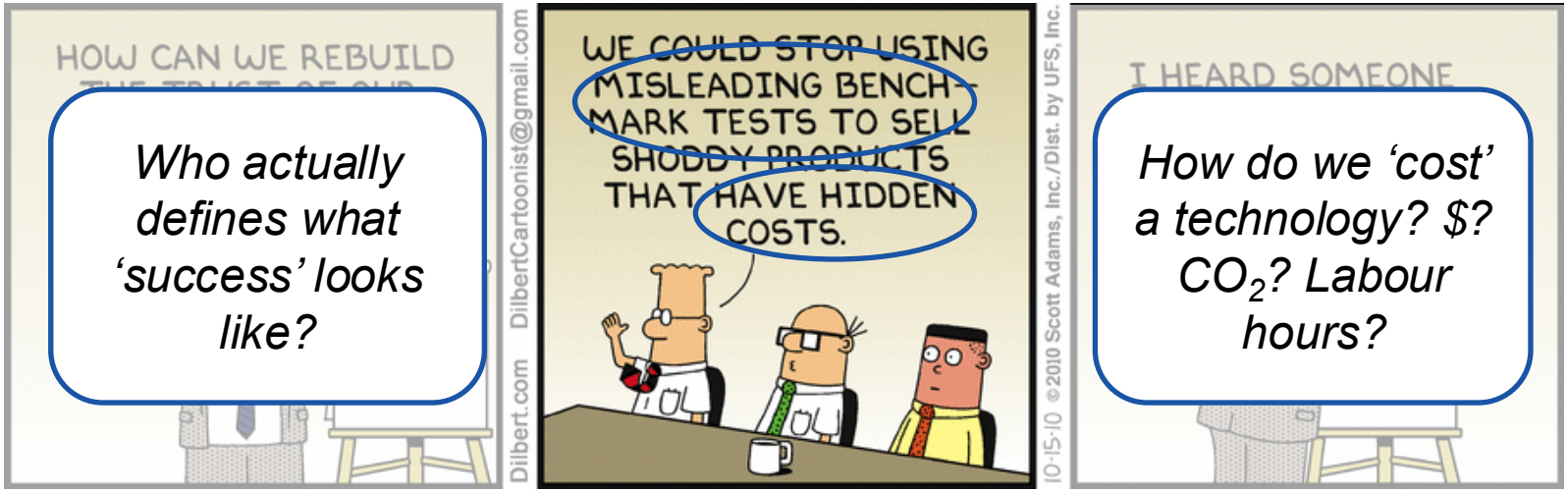
Embracing Ethics as an (AI) Engineer

A (cynical) take on how it feels some* perceive the notion of ethical AI…

Embracing Ethics as an (AI) Engineer



Embracing Ethics as an (AI) Engineer



Embracing Ethics as an (AI) Engineer

Even for those of us who are on-board with pursuing responsible innovation… it can feel like a restrictive, ‘tick-box’ exercise where we just need to take our existing research/development plan and somehow adjust or ‘rein it in’ to make it pass some pre-provided set of rules or guidelines.

Embracing Ethics as an (AI) Engineer

Instead, from the very beginning of a new project, let's:

- Design, deploy and evaluate new tech. using *inherently responsible and ethical* approaches
- Consider the *ethical implications* and any *associated risk* from our tech.
- Critically evaluate the *real-world* impact(s) (good *and bad*) of our tech.

Embracing Ethics as an (AI) Engineer

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because (i) this leads to *better* [smarter / more impactful / more innovative] technologies, *whilst also* allowing us to follow the 'do no harm' principle.

Embracing Ethics as an (AI) Engineer

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- Critically evaluate the *real-world* impact(s) (good *and bad*) of our tech.

and (ii) we have something to offer here! We are good at analysing and trying to solve complex problems! This is an opportunity to creatively apply those skills in considering more 'social' constraints and outcomes.

How?

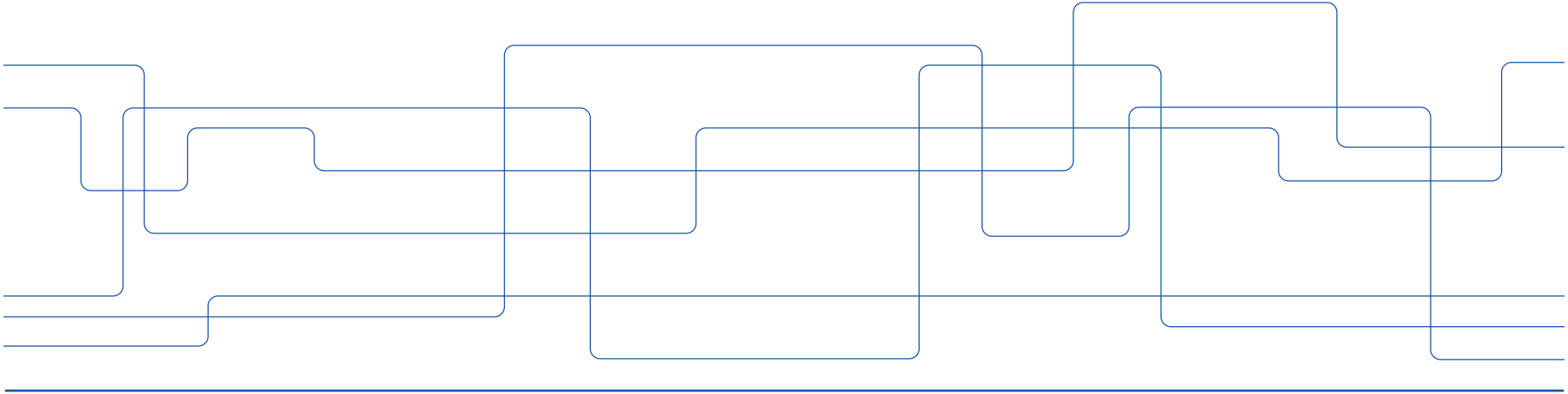
Two practical things you can do, from today, to deliver more ethical AI:

1. Taking a mutual shaping view of, and approach to, technology and society.

2. Using ethical standards for best practice and ethical risk assessment.

I'll be asking you to apply these to a worked example later ;)

Mutual Shaping of Technology and Society



Mutual Shaping of Technology & Society

Šabanović, Selma. "Robots in society, society in robots." *International Journal of Social Robotics* 2.4 (2010): 439-450.

[available here and on canvas: https://homes.luddy.indiana.edu/selmas/Sabanovic-IJSR2010.pdf]



Robots in Society, Society in Robots

Mutual Shaping of Society and Technology as a Framework for Social Robot Design

Mutual Shaping of Technology & Society

The prevalent world view in robotics and AI has traditionally been *technological determinism,* which can be summarised with the saying:

"Science finds ~ industry applies ~ man conforms"

Typical signs of **technological determinism**:

- Technological advancement ≡ societal success
- Focus on (justifying) technological *fixes,* not understanding the *problem*
- Design/application choices being made only by 'technical' experts

Mutual Shaping of Technology & Society

TD is a pretty arrogant and narrow viewpoint:

*"In their **emphasis on technological advancement as the determinant of societal success, roboticists define the societal issues they wish to solve** 'not [as] social, political, cultural, historical problems, but [as] biotechnological problems that call for biotechnological solutions'. **Society, in turn, is displayed as a passive receptor of the products** of visioning and negotiation performed by experts in the process of design."*

TD neglects human agency and ignores social context

Mutual Shaping of Technology & Society

Luckily for us, Sabanovic then provides us with some examples of iterative, contextual, and participatory design and evaluation practices that:

“enable the mutual shaping of technology and society to play out through iteration between social analysis and technology design.”

[1] Situating Technology Design in Society

[1A] Evaluate technology in society

- Get out of the lab and test things out '*in the wild*'; in the environments they're expected to go in and with the people who are actually expected to use them

[1] Situating Technology Design in Society

[1A] Evaluate technology in society

- Get out of the lab and test things out ‘in the wild’; in the environments they’re expected to go in and with the people who are actually expected to use them

[1B] Studying socio-technological ecologies

- Undertake systemic study of the context of use *before* starting to design and build things! *and* again once the tech. has been deployed to look at performance, how it’s being used and the impact it’s had on the socio-technical ecology

Example: Interactions with Roomba…



Figure 1. The Hoover Flair stick vacuum (left) and the i-Robot Roomba Discovery vaccum (right) used in the study.

- 6 families, initial interviews to understand household management
- Half given a Roomba and half given an equivalent traditional vacuum
- Photo diary to capture first 12 uses
- Follow-up interviews at 3, 6, 9 and 12 months

Forlizzi, Jodi. "How robotic products become social products: an ethnographic study of cleaning in the home." *2007 2nd ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 2007.

…the Pet-Friendly, Feminist Robot Vacuum?





“They also saw an evolution of the social system of cleaning after the Roomba came into the house—household members, including teenagers and males, who had not been participating in cleaning activities started to do so when the Roomba was involved.”

Sabanovic on Forlizzi’s. "How robotic products become social products: an ethnographic study of cleaning in the home."

…the Pet-Friendly, Feminist Robot Vacuum?





“They also saw an evolution of the social system of cleaning after the Roomba came into the house—household members, including teenagers and males, who had not been participating in cleaning activities started to do so when the Roomba was involved.”

Sabanovic on Forlizzi’s. "How robotic products become social products: an ethnographic study of cleaning in the home."

[1] Situating Technology Design in Society

[1C] Outside-in Design

- Combines 1A (evaluating technology in society) and 1B (studying socio-technological ecologies) into an iterative cycle.

  Sabanovic says: *“It starts with situated observation and empirical research and follows with an iterative process of prototype building and testing in the real world to construct robots through a continuing conversation with the social environment of their use”*

[1] Situating Technology Design in Society

A nice thought with which Sabanovic concludes on this is that such an approach can encourage roboticists and technologists like us to think of ourselves: *“not as primarily technologists, solving technical problems, but as* ***problem-oriented researchers who can work across disciplinary boundaries and consciously tackle social issues****”.*

[2] Involving Society in Technology Design

Including users at the early stages of design is important for *“developing socially robust, rather than merely acceptable, robotic technologies.”*

Stakeholders (people who are going to be directly impacted by the tech. being developed, and who have tacit knowledge of the application domain) should have opportunities to influence its design, the kind of applications to be developed, how such tech. will be used and what their broader social meaning will be.

Participatory design, co-design: *focus groups, interviews, workshops, interdisciplinary hackathons, crowdsourcing…*

Example: A Study with Therapists

Focus Group Structure	
Pre-Discussion	Acceptability questionnaire
Discussion Part 1	Conventional therapy delivery Using social robots for therapy
Project Talk & Demo	Project motivations Supporting literature 2x robot demonstrations
Discussion Part 2	Revisit of social robots for therapy
Post-Discussion	Acceptability questionnaire

Winkle, Katie, et al. "Mutual shaping in the design of socially assistive robots: A case study on social robots for therapy." *International Journal of Social Robotics* (2019): 1-20.

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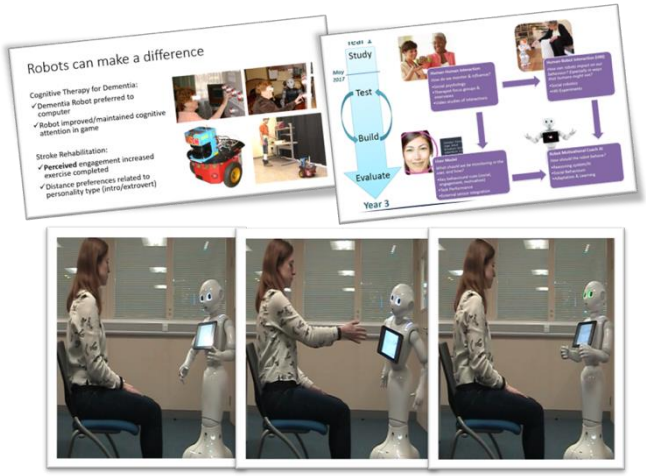
Mutual Learning 1
Participants → Researcher

- share expertise + understanding of current use context
- give own ideas before being biased

Winkle, Katie, et al. "Mutual shaping in the design of socially assistive robots: A case study on social robots for therapy." *International Journal of Social Robotics* (2019): 1-20.

Example: A Study with Therapists

Focus Group Structure	
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Post-Discussion	Acceptability questionnaire



Mutual Learning 2
Researcher → Participants

- share motivations
- showcase technology

Winkle, Katie, et al. "Mutual shaping in the design of socially assistive robots: A case study on social robots for therapy." *International Journal of Social Robotics* (2019): 1-20.

Example: A Study with Therapists

Focus Group Structure	
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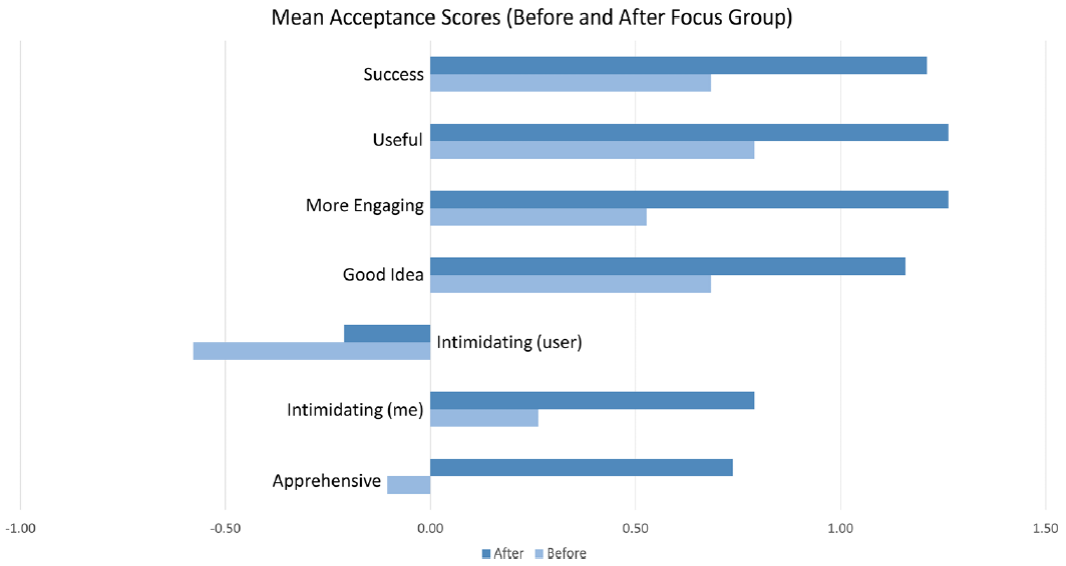
Mutual Shaping
Researcher ←→ Participants

- revisit discussions as equal participants
- ensures data of interest (i.e. that accounts for tech. constraints) still collected

Winkle, Katie, et al. "Mutual shaping in the design of socially assistive robots: A case study on social robots for therapy." *International Journal of Social Robotics* (2019): 1-20.

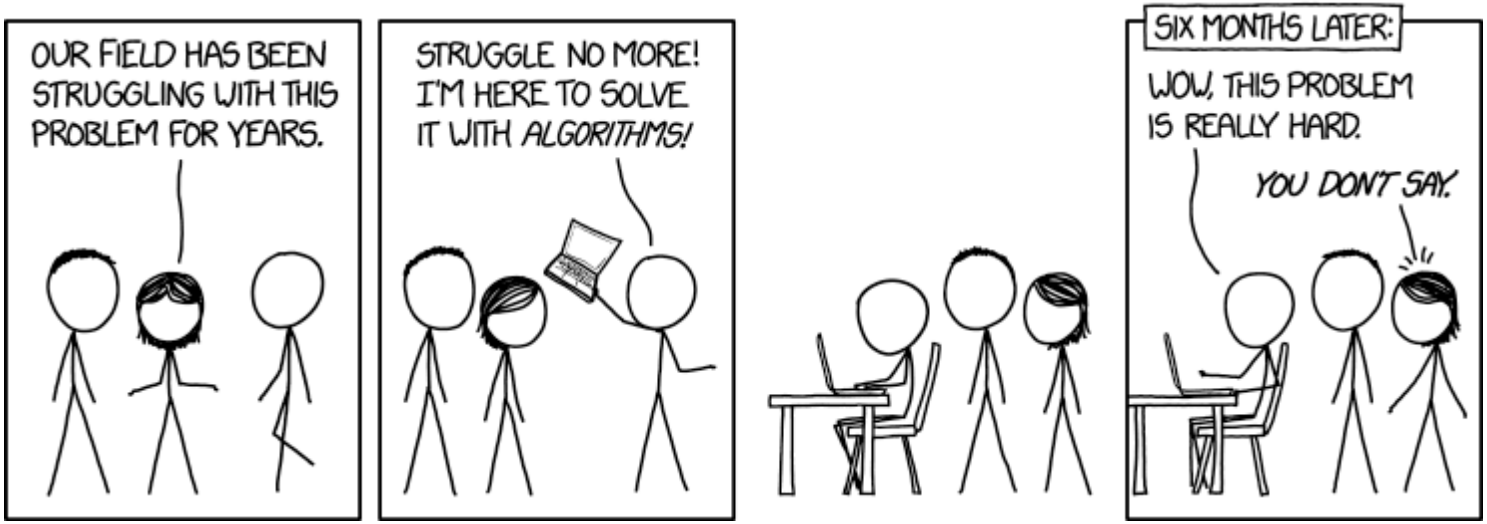
Example: A Study with Therapists

- New ideas for applications as well as the Pepper/project-specific) data required
- Identified real world considerations academically interesting & crucial to acceptability
- Significantly increased therapists' own acceptance of robots



Winkle, Katie, et al. "Mutual shaping in the design of socially assistive robots: A case study on social robots for therapy." *International Journal of Social Robotics* (2019): 1-20.

TL; DR: Mutual Shaping helps to avoid…



Part 1 Summary

We talked through some topical issues around the Ethics of AI – energy and exploitation, anthropomorphism and deception, racism and sexism. But we also identified our working definition of *Ethical AI* to be '*any and all AI designed ethically'* – with a focus on how we can develop new technologies following the minimum guiding principle of *do no harm.*

Part 1 Summary

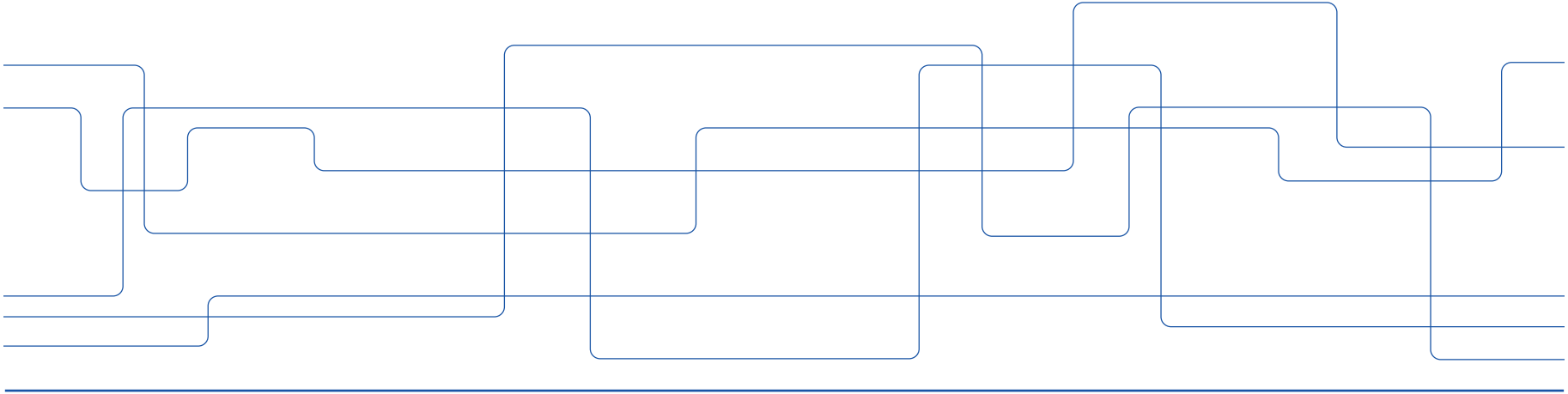
I suggested that maybe this concept of ‘being ethical’ gets a bit of a tough time in the tech. community (whether that’s in industry or academia) because it can feel like a bit of a chore that might somehow restrain our work.

But a key purpose of today is for me to try and show you how, on the contrary, this leads to better tech *and* is pretty interesting to work on and consider.

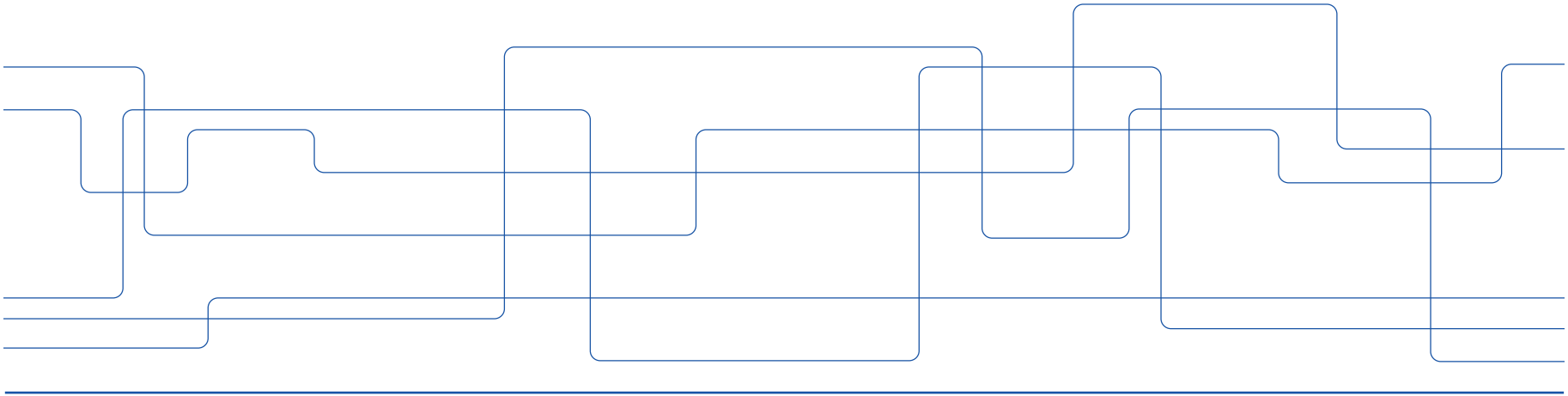
Part 1 Summary

Finally, we introduced the concept of *mutual shaping,* both as a (better) way to think about the interactions between technology and society, but also as an approach to take when developing new tech. Practically this means doing a *thorough* study of the 'problem' (in all it's socio-technological complexity) and pursuing participatory design and evaluation methodologies with domain experts, users and other stakeholders.

Time for a Break!



Ethical Standards in AI and Robotics



Ethical Standards in AI and Robotics

Quick acknowledgement! ~

The following draws a lot on the work of Prof. Alan Winfield on the (ongoing) ROBOTIPS project.

Alan was the first ever Professor of Robot Ethics (that we know of) and has heavily contributed to national and international initiatives on ethical robot standards.

Check out his blog (https://alanwinfield.blogspot.com/) for robots, ethics and more.



Bristol Robotics Laboratory

RoboTIPS
RESPONSIBLE ROBOTS FOR THE DIGITAL ECONOMY

What are Standards?



Standards vs Principles

An informal survey in 2019 revealed a total of 25 different sets of ethical *principles* for robotics and artificial intelligence (AI) to date.

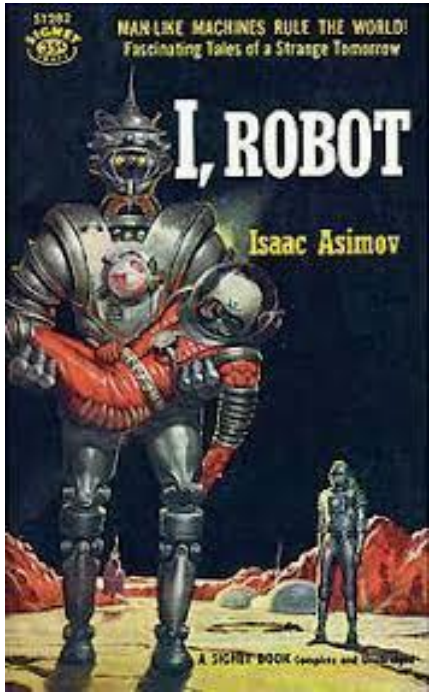
The earliest (1950) are Asimov's laws of robotics: important because they established the principle that robots should be governed by principles.

But a lot of them are more recent - 22 were published from 2017 onwards

https://alanwinfield.blogspot.com/2019/04/an-updated-round-up-of-ethical.html

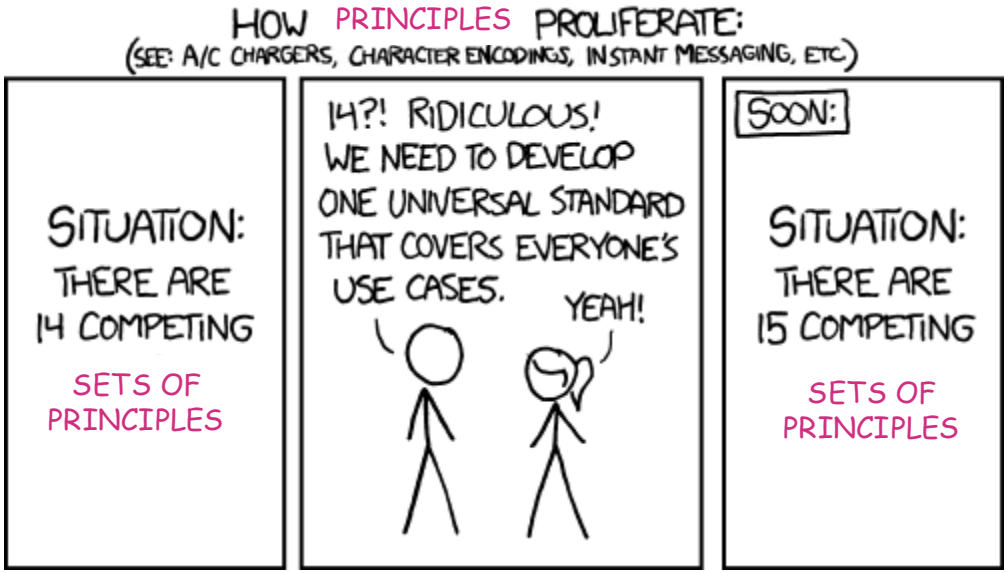
Asimov’s three laws of Robotics (1950)

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.



Standards vs Principles

A cynical person might say it's much easier to *publish* another set of principles than it is to actually *implement* them:



Standards vs Principles

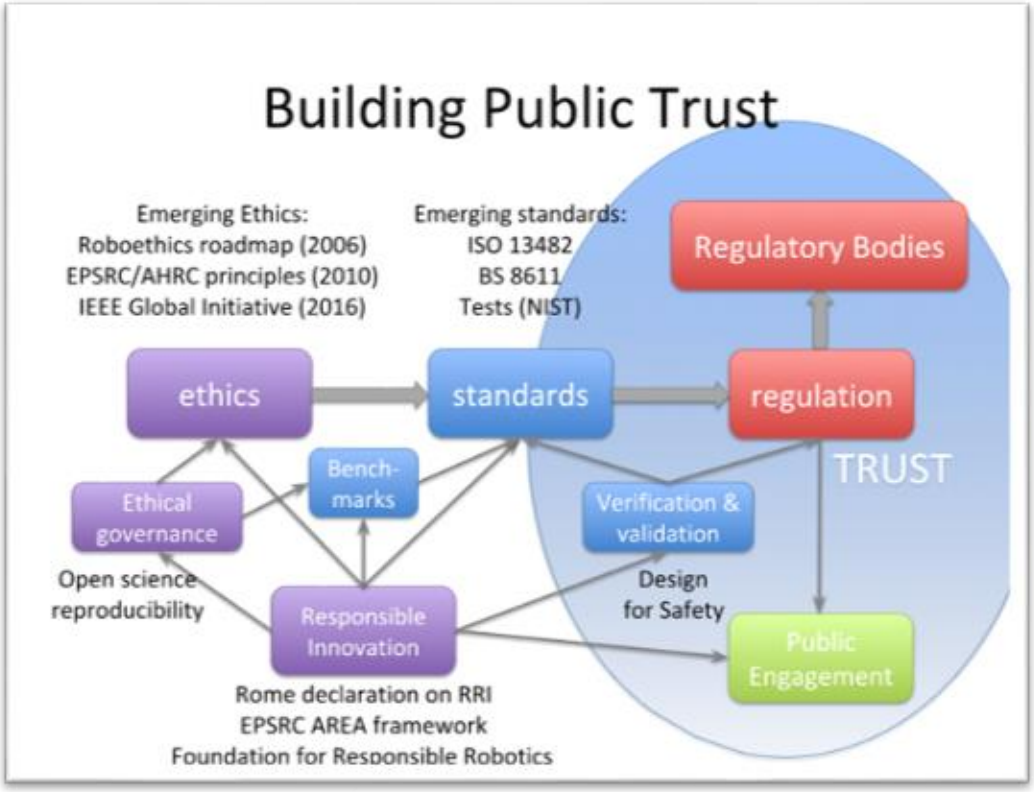
Principles are not practice.

Standards are vital in **bridging the gap between good intentions and good practice**, as demonstrated by current big-impact industries.

Standards, Regulation and Trust



Ethics – Standards – Regulation => Trust



Ethics – Standards – Regulation => Trust



Ethics – Standards – Regulation => Trust

Some key initiatives (and practical resources) to be aware of. These can help you assess and critique new technologies/typical design and development methods you observe as well as inform your own practice.

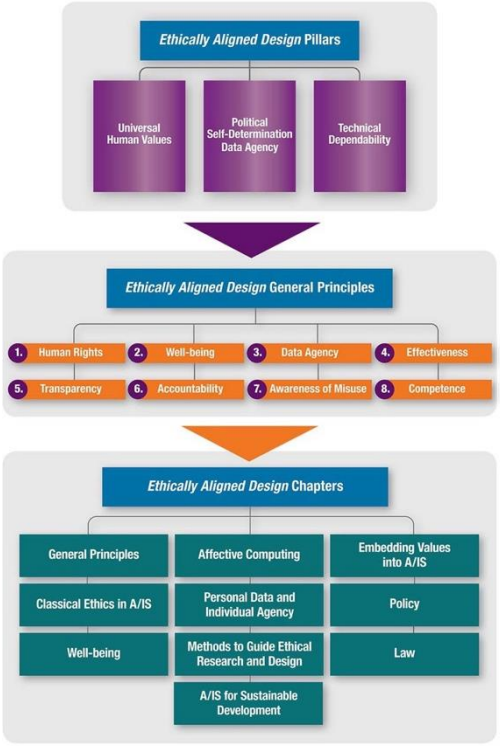
IEEE Ethics in Action



"The goal of The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems ("The IEEE Global Initiative") is that Ethically Aligned Design will ***provide pragmatic and directional insights and recommendations, serving as a key reference for the work of technologists, educators and policymakers*** *in the coming years."*

https://standards.ieee.org/content/dam/ieee-standards/standards/web/documents/other/ead1e.pdf

IEEE Ethics in Action



EAD1e: *"provides a mapping of the conceptual framework of Ethically Aligned Design. It outlines the logic behind "Three Pillars" that form the basis of EAD1e, and it connects the Pillars to high-level "General Principles" which guide all manner of ethical A/IS design."*

This is also informing current work on a bunch of new *standards* (although EAD1e itself does provide some recommendations on putting this stuff into practice)

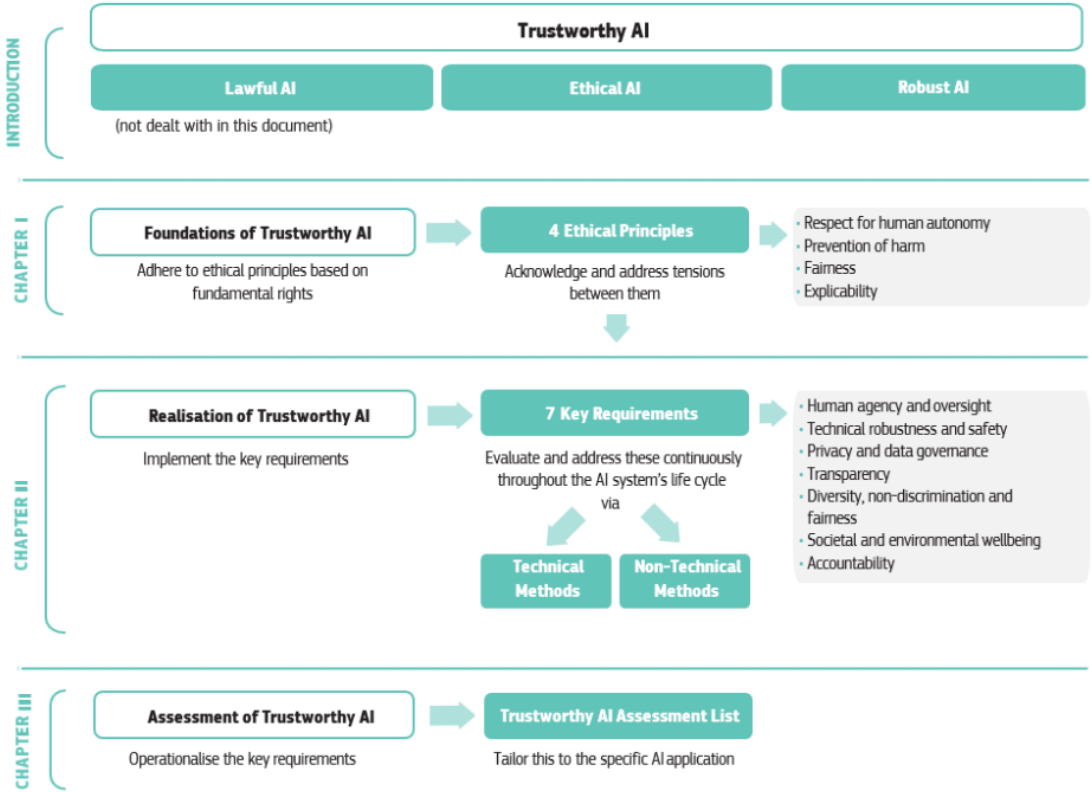
IEEE Ethics in Action

Standard	Remit
P7000	Model Process for Addressing Ethical Concerns During System Design
P7001	Transparency of Autonomous Systems
P7002	Data Privacy Process
P7003	Algorithmic Bias Considerations
P7004	Standard on Child and Student Data Governance
P7005	Standard on Employer Data Governance
P7006	Standard on Personal Data AI Agent Working Group
P7007	Ontological Standard for Ethically driven Robotics and Automation Systems
P7008	Standard for Ethically Driven Nudging for Robotic, Intelligent and Autonomous Systems
P7009	Standard for Fail-Safe Design of Autonomous and Semi-Autonomous Systems
IEEE Std 7010™-2020	Recommended Practice for Assessing the Impact of Autonomous and Intelligent Systems on Human Well-Being
P7011	Process of Identifying & Rating the Trust-worthiness of News Sources
P7012	Standard for Machine Readable Personal Privacy Terms
P7014	Ethical considerations in Emulated Empathy in Autonomous and Intelligent Systems

- 14 standards, many are now published, some are still being worked on by expert working groups (that you can join!)

- Standards exist; the question is adoption…

European Commission
Ethics Guidelines for Trustworthy AI



https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

TL; DR: The EU AI Act

- The EU aims to regulate AI to ensure its *safe* and *beneficial* use
- AI systems will be classified based on their risk levels, with different regulations for each
  - Unacceptable risk AI systems, like those used for social scoring or harmful manipulation, will be banned
  - High-risk AI systems, used in critical areas like healthcare or law enforcement, will be strictly regulated
- Transparency requirements will be enforced for generative AI models like ChatGPT
- The AI Act expected to “begin” in 2026, but with some delays: https://www.gibsondunn.com/eu-ai-act-omnibus-agreement-postponed-high-risk-deadlines-and-other-key-changes/

https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence

Key Takeaways on Ethical Guidelines and Regulation

- The Ethics Guidelines are recommendations, while the AI Act is a legally binding regulation with enforcement mechanisms
- The AI Act's risk-based approach categorizes AI systems based on their potential impact, with stricter regulations for high-risk systems

Both instruments emphasize a human-centric approach to AI, prioritizing human safety, well-being and fundamental rights.

ISO 42001: **AI governance and compliance**

1. What ISO/IEC 42001 is

ISO/IEC 42001 is the International Standard for AI management systems.

It provides requirements and guidance **for organizations that develop, provide or use AI systems**. The standard helps organizations **manage risks related to AI** while supporting innovation, trust and accountability.

ISO/IEC 42001 is the first global standard that defines how to establish, implement, maintain and continually improve an AI management system.

2. What an AI management system means

An AI management system is a structured set of policies, processes and controls that help organizations govern how AI systems are designed, developed, deployed and used.

In practice, an AI management system helps organizations:

- define responsibilities for AI use
- identify and assess AI-related risks
- ensure transparency and accountability
- manage data quality and system performance
- address ethical, legal and societal concerns
- monitor AI systems throughout their lifecycle

https://www.iso.org/home/insights-news/resources/iso-42001-explained-what-it-is.html

BS8611: Ethical Risk Assessment

British Standard BS8611 articulates a set of 20 distinct ethical hazards and risks grouped under the categories:

- Societal Risks
- Application Risks
- Commercial/Financial Risks
- Environmental Risks

and provides advice for mitigating the impact of each risk, plus suggestions on how such measures might be verified or validated.



Updated 2023 version:
https://knowledge.bsigroup.com/products/robots-and-robotic-devices-ethical-design-and-application-of-robots-and-robotic-systems-guide

Example: An ERA of RoboTed

RoboTed uses the following technologies:

- Internet (WiFi) connection
- Cloud-based speech recognition and conversational AI (chatbot) plus local speech synthesis
- Camera eyes for (local) facial recognition
- Motorised arms and legs to provide for limited baby-like movement and locomotion.



A. Winfield and K. Winkle, “RoboTed: a case study in Ethical Risk Assessment" in Proceedings of the 2020 International Conference on Robot Ethics and Standards [available here]

Example: An ERA of RoboTed

RoboTed is designed to:

- Recognise its owner, turning its face toward them
- Respond to physical play such as hugs and tickles.
- Tell stories (allowing for interruptions to answer questions/repeat sections)
- Sing songs and encourage singing along
- Act as a child minder, allowing parents to both remotely listen, watch and speak via RoboTed.



A. Winfield and K. Winkle, “RoboTed: a case study in Ethical Risk Assessment" in Proceedings of the 2020 International Conference on Robot Ethics and Standards [available here]

Example: An ERA of RoboTed

Table 1. Physical Risk Analysis for RoboTed

Hazard	Risk	Level[a]	Mitigation
Tripping	User(s) trip over RoboTed when it is crawling on the floor	M	Audible crawling sound to alert users to its presence
Battery overheating	Defective batteries or battery chargers can overheat or *in extremis* catch fire	M	Design to make use of consumer rechargeable batteries rather than high-risk Li-Ion batteries.

Note: [a]Risk level: (H)igh, (M)edium or (L)ow



A. Winfield and K. Winkle, “RoboTed: a case study in Ethical Risk Assessment" in Proceedings of the 2020 International Conference on Robot Ethics and Standards [available here]

Example: An ERA of RoboTed

Table 2. Psychological Risk Analysis for RoboTed

Hazard	Risk	Level[a]	Mitigation
Addiction	Child plays with RoboTed obsessively and neglects family	M	Explore 'RoboTed needs to sleep now' function
Deception (of child)	Child believes that RoboTed has feelings (for her)	M	Design chatbot to avoid language that suggests feelings
Over trusting (by child)	Child cannot distinguish mode of operation	H	Notification when child minder mode activated, uses parents' voice rather than RoboTed's voice
Over trusting (by parents)	Parents come to rely on the child minder function	H	Remove the child minder function
The Uncanny Valley	Child becomes fearful of robot	L	Use cartoon voice; engage children in early user trials

Note: [a](H)igh, (M)edium or (L)ow



A. Winfield and K. Winkle, "RoboTed: a case study in Ethical Risk Assessment" in Proceedings of the 2020 International Conference on Robot Ethics and Standards [available here]

Example: An ERA of RoboTed

Table 3. Privacy and Security Risk Analysis for RoboTed

Hazard	Risk	Level[a]	Mitigation
Weak security	Malicious hackers gain access to RoboTeds sensors & control functions	H	Implement strong encryption together with best practice password protection
Privacy	Personal data, including images and voice recordings of child are stolen	M	Put in place auditable measures to ensure personal data is deleted immediately
Lack of transparency	Lack of data logs makes it hard or impossible to investigate accidents	H	Build a secure local data logger into RoboTed

Note: [a] (H)igh, (M)edium or (L)ow



A. Winfield and K. Winkle, “RoboTed: a case study in Ethical Risk Assessment" in Proceedings of the 2020 International Conference on Robot Ethics and Standards [available here]

Example: An ERA of RoboTed

Table 4. Environmental Risk Analysis for RoboTed

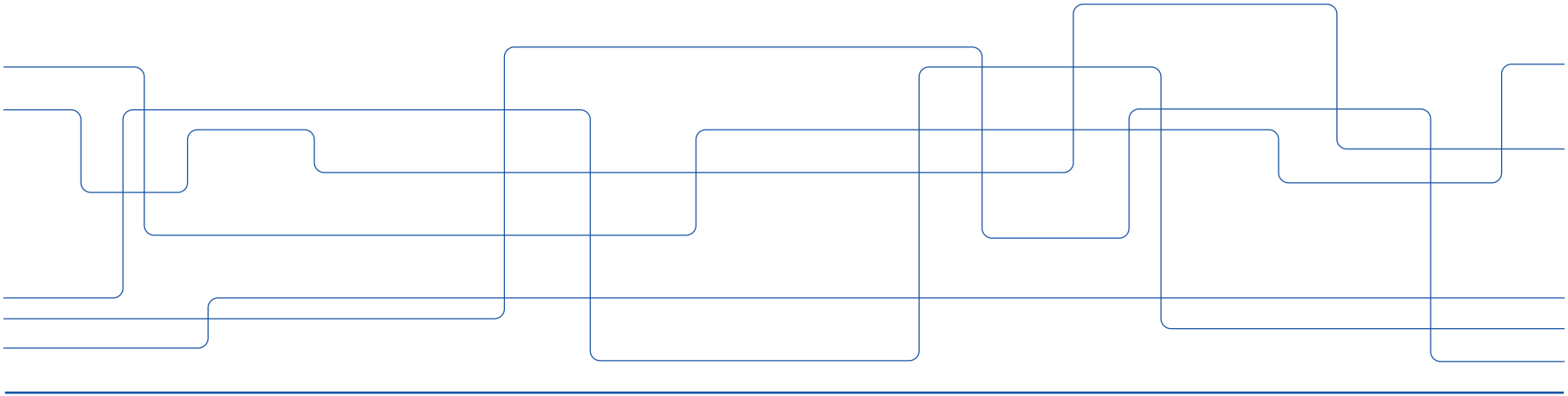
Hazard	Risk	Level[a]	Mitigation
Unsustainability (of materials)	Robot uses unsustainable or high carbon cost materials	M	Use materials (e.g. RoboTeds fur) from sustainable sources, avoiding plastics.
Unrepairability	Robots lifetime is limited because faults cannot be repaired or parts replaced	M	Design for ease of repair with replaceable parts – especially battery
Unrecyclability	End of life robots are dumped in land fill	M	Design for ease of recycling parts and materials

Note: [a] (H)igh, (M)edium or (L)ow



A. Winfield and K. Winkle, “RoboTed: a case study in Ethical Risk Assessment" in Proceedings of the 2020 International Conference on Robot Ethics and Standards [available here]

Worked Example – Your Turn



Brief: Autonomous Robot Fitness Coach

You are tasked to design an autonomous social robot to help motivate people through a pre-prescribed national health service programme. The aim is to improve motivation in sessions where people would normally work out alone.

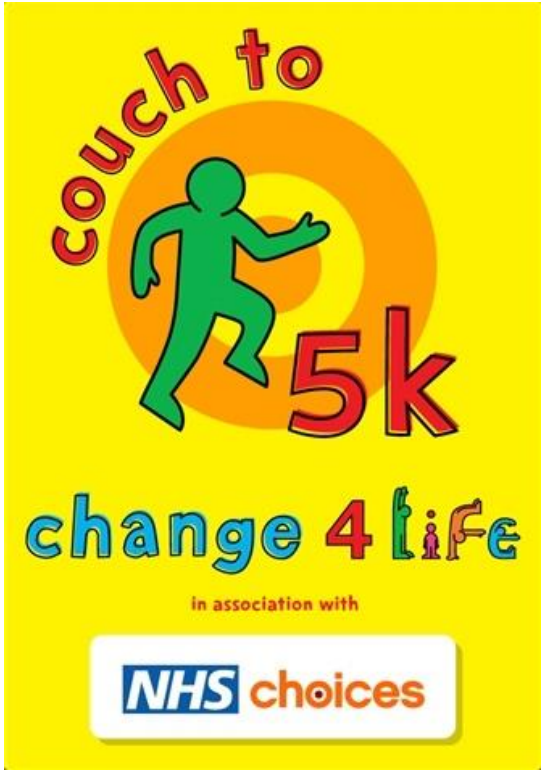




Left:: J. Fasola and M. J. Matarić, 'Robot exercise instructor: a socially assistive robot system to monitor and encourage physical exercise for the elderly', in RO-MAN, 2010 IEEE, 2010, pp. 416–421.

Right: S. Schneider, M. Goerlich, and F. Kummert, 'A framework for designing socially assistive robot interactions', Cognitive Systems Research, vol. 43, no. Supplement C, pp. 301–312, Jun. 2017.

Exercise Programme: NHS Couch to 5km



Early sessions include short burst of running interspersed with periods of walking (e.g. 1 minute walk, 1.5 minute run x 8).

Running times get longer and longer until you run non-stop for about 30 minutes/cover about 5km. The whole programme is 27 sessions, designed to be undertaken as 3x sessions per week for 9 weeks.

The programme is currently delivered by an app, and is designed for people of all ages who have no major health concerns that prevent safely running (slowly if necessary) for about 30 minutes non-stop.

Task 1: Taking a Mutual Shaping Approach

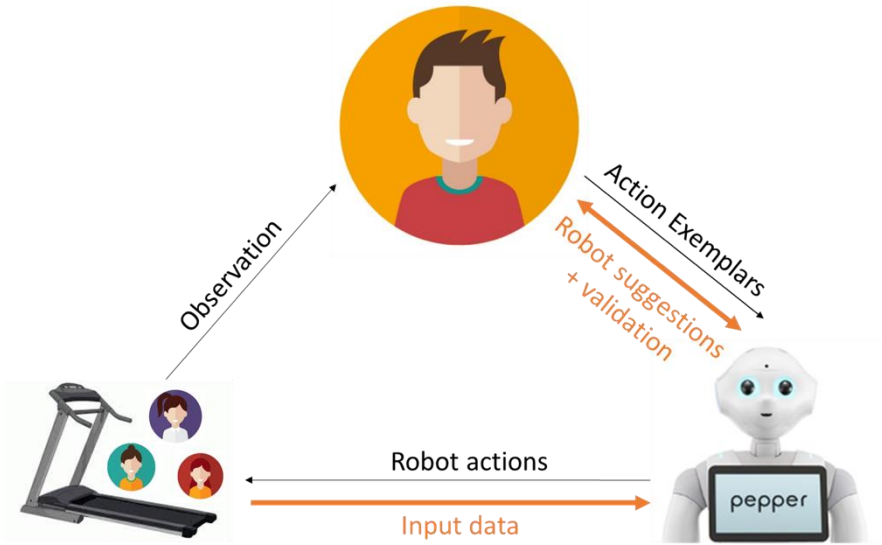
Can you sketch out a research and development pipeline, detailing the key activities you would undertake between getting this brief and delivering the finished, ready-to-use robot?

Task 1: Taking a Mutual Shaping Approach

Can you sketch out a research and development pipeline, detailing the key activities you would undertake between getting this brief and delivering the finished, ready-to-use robot?

- What are the key design, development and evaluation activities you might want to do? Some key milestones?
- Who would you want to talk to? Anywhere you want to visit? When?
- Any thoughts on how to automate the robot, and any related mutual shaping considerations to be made?

Expert-in-the-Loop IML as Participatory Design

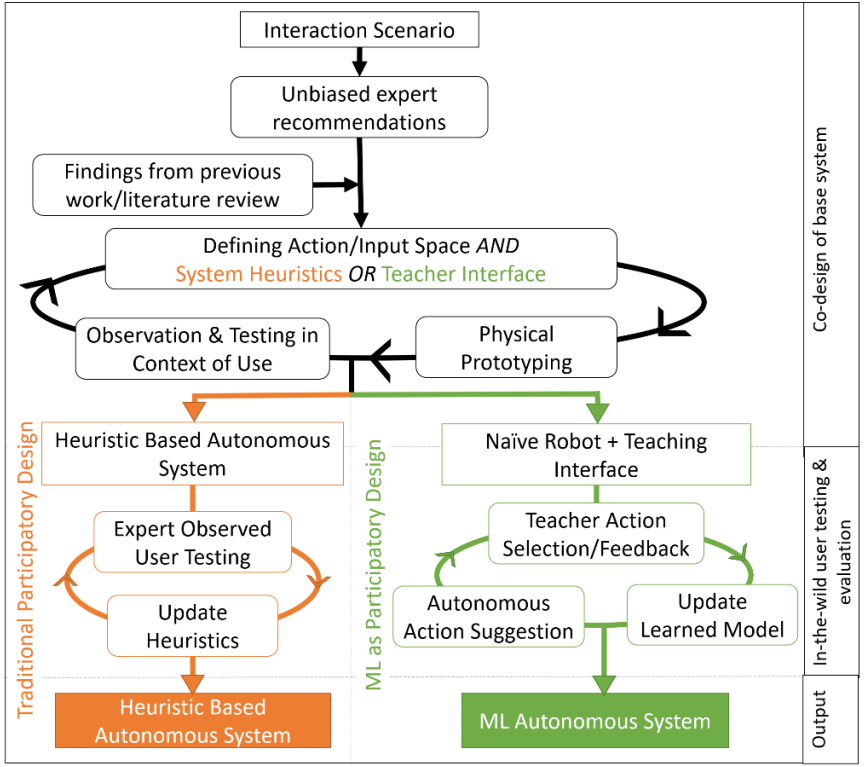


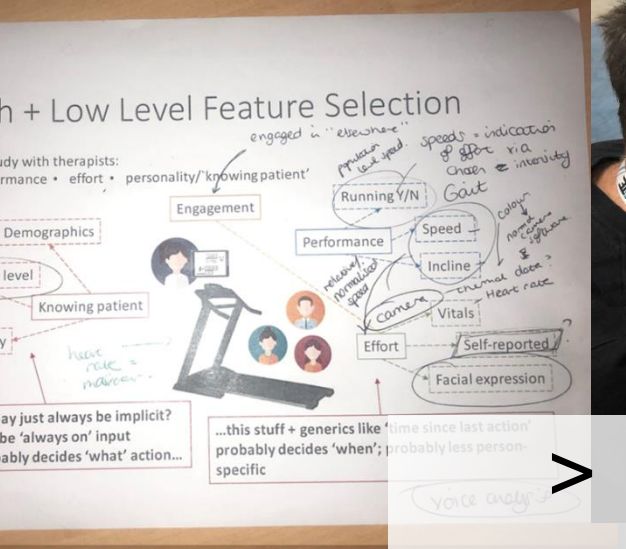
1. Co-design robot actions and input space with a *domain* expert

2. Co-design a ‘teaching interface’ for using those actions and responding to robot suggestions

3. Domain expert *teaches* the robot via interactive machine learning in-the-wild

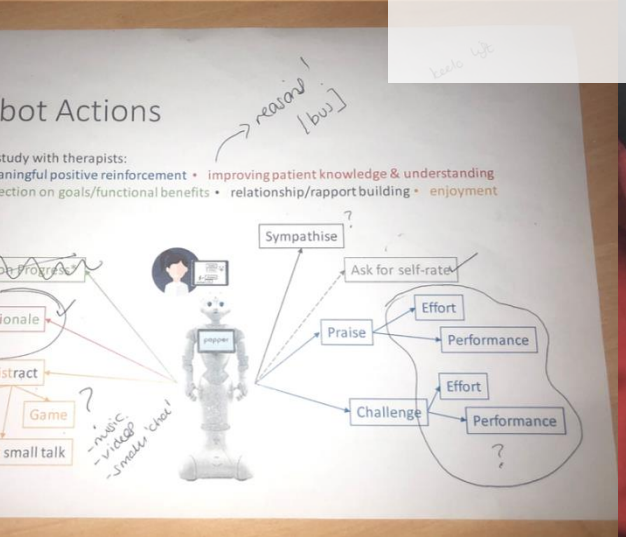
Expert-in-the-Loop IML as Participatory Design















> 12 hours co-design sessions

Expert-in-the-Loop IML as Participatory Design

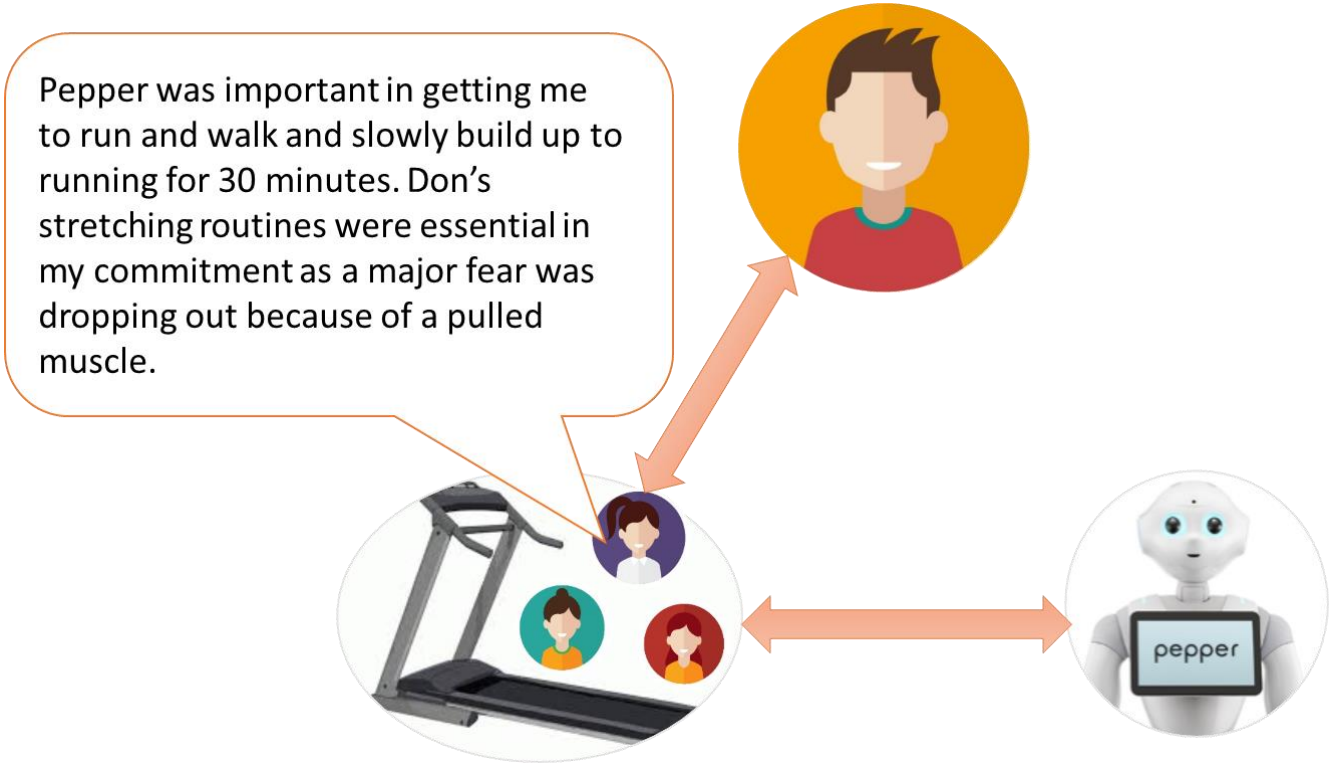
Installed in university gym for 3 months, used with 10 participants of whom 1 dropped out

Ran a total of 232 exercise sessions

- 151 supervised (training the ML)
- 32 autonomous, trained ML
- 49 autonomous heuristic based ‘control’



Expert-in-the-Loop IML as Participatory Design



Expert-in-the-Loop IML as Participatory Design

- ✓ Studied context of use before deploying - with some willing colleagues as test subjects ;)
- ✓ Worked with a domain expert throughout design *and* automation
- ✓ Domain expert presence throughout ensured safety of participants
- ✓ Final system ‘explainable’ in the sense of knowing exactly where the training data has come from
- ✓ Real world deployment allowed for study of emergent mutual shaping effects

…something obvious they didn’t do, but ideally would have?

Expert-in-the-Loop IML as Participatory Design

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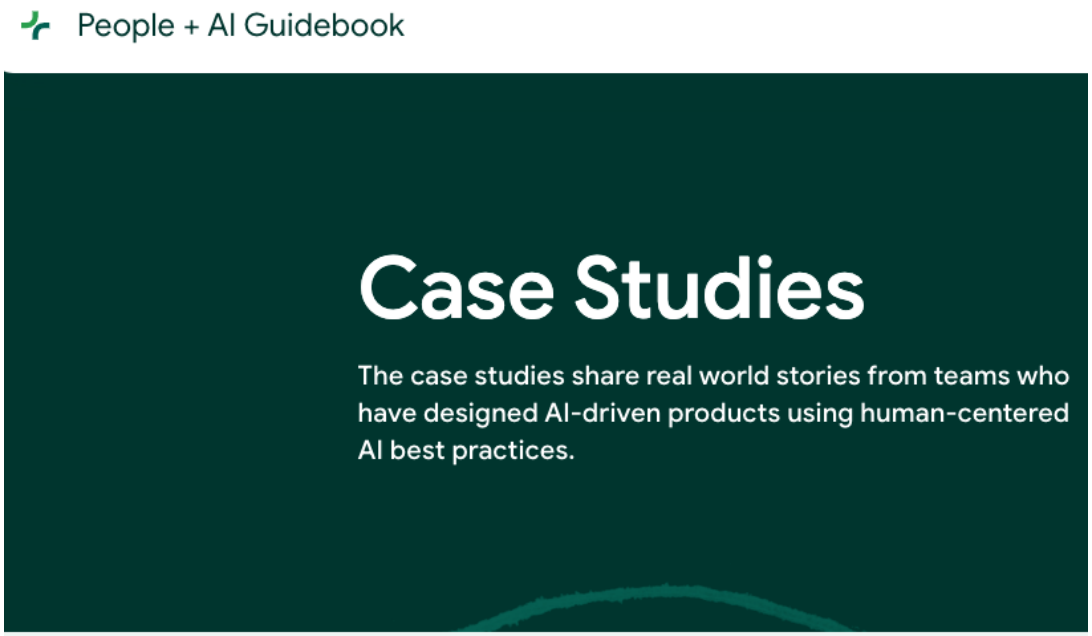
X No design/development work with *target users*

Task 2: Ethical Risk Assessment



With reference to BS8611, IEEE's EAD1e or by taking inspiration from the RoboTed example, can you identify some ethical risks that might arise from our robot fitness coach? Can you also suggest any potential mitigating strategies?

More case studies



https://pair.withgoogle.com/guidebook/case-studies

Final Recap

Technological determinism has driven a lot of tech. development, and likely resulted in a lot the problematic issues we see know (e.g. racism, sexism in AI).

A more accurate view of tech in the world can be achieved by taking a mutual shaping view. Actively pursuing mutual shaping means to (i) properly situate tech into our lives and social systems and (ii) invite society into tech design. Both of which leads to more effective tech.

Emerging ethical standards build on this provide *practical guidance* for ethical AI development and *tangible* ways to assess and address ethical risk.

Final Recap

AI can impact several aspects of our daily lives, such as

- Labour
- Surveillance & privacy
- Sustainability
- Decision-making (credit-score attribution, health diagnosis, recommendations, …)
- …

Take Home Message

Ethical AI (that is *any* AI that is designed, developed and deployed ethically) is *better* AI, and is incredibly fulfilling to develop.

Assignment: Essay on Ethical and Societal Aspects of AI

Step 0

- Form a group of 3-4 people (4 is strongly preferred).
- Register as a group on Canvas
- Agree on a meeting date

Step 1 (individually)

- Choose at least two papers to read (from the ones listed in the assignment on canvas)
- Write an essay (500-600 words excluding references) summarizing the most important take-aways regarding ethical and societal aspects of AI from the reading material

Assignment: Essay on Ethical and Societal Aspects of AI

Step 2

- Meet with your essay group and discuss your findings from individual reflection
- Take a picture! ☺
- Produce a text (500-600 words) that summarizes the conclusions from this discussion

Deadline

- **Nov 9, 7:00 pm**

Read the assignment description carefully on canvas for all the details, particularly the note on using LLMs in this assignment.